

Project Title

AI-Powered Identity Verification and Fraud Prevention Using UID Aadhaar

Abstract

This research document outlines a practical, ethical, and technically feasible approach to build an AI-powered fraud management and identity verification system that leverages Aadhaar (UID) and other KYC documents. The system combines AI-OCR, deep learning-based forgery detection, computer vision, and Azure cloud services (Azure AI Document Intelligence and Azure OpenAI) to automate document extraction, validate document authenticity, detect tampering, and integrate into KYC workflows for financial institutions.

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1. Introduction and Motivation

Digital onboarding for banking, fintech, and other services requires fast and accurate identity verification. Aadhaar is a primary identity document in India and is often used in KYC pipelines. Manual verification is time-consuming and error-prone; automated AI solutions can speed up onboarding and reduce fraud. This project focuses on building models and system components to extract structured data from Aadhaar and related documents, detect forged or tampered documents, and generate real-time risk flags integrated into existing KYC platforms.

2. Objectives and Expected Outcomes

Primary objectives

- Build a robust AI-OCR pipeline to extract fields from Aadhaar and other KYC documents (PAN, passport, driving license).
- Develop deep learning models to detect image-level tampering, photo overlays, font inconsistencies, and mismatches between extracted fields and claimed identity.
- Integrate models into a test KYC workflow, deployed on Azure for low-latency inference.

Expected outcomes

- Automated extraction of key fields (name, Aadhaar number masked/unmasked in safe testing, DOB, address, photo) with per-field confidence scores.
- A fraud-risk scoring module that flags documents exhibiting signs of tampering, spoofing, or inconsistency.
- Sample UI for operations team to review the flagged cases with highlighted evidence (cropped regions, mismatch reports).

3. Datasets (open-source) — required and alternative sources

Note: Aadhaar is sensitive personal data. Use only properly consented, anonymized, or synthetic datasets for training and testing. For real production use, follow legal and regulatory rules and obtain necessary approvals.

Primary dataset provided by user

- [AADHAR_DATASET](#) on Kaggle (user: nagendra048) — image samples that can be used for OCR/model development and prototyping. (Kaggle dataset identifier: [nagendra048/aadhar-dataset](#)).

Additional open-source datasets & resources to consider

- Kaggle: Aadhaar images, Aadhaar-like datasets, PAN card datasets (examples: [varunkumargera/aadhar-images](#), [nagendra048/pan-card-dataset](#), [sparsh2002/govtiddataset](#)).
- Roboflow public datasets for Aadhaar-like card detection (useful for bounding-box training and detection).
- Government sources and synthetic data resources: UIDAI (Offline eKYC sample data) and Open Government Data ([data.gov.in](#) Aadhaar group) for non-sensitive, aggregated datasets and sample formats.
- Synthetic data generation: use scripted templates to create synthetic Aadhaar-like images (masking real numbers) and data augmentation to simulate wear/blur/photocopy effects.

Required dataset types for the project

1. Aadhaar-front/back images with bounding-box labels for fields (name, Aadhaar number, DOB, gender, photo).
2. PAN passport & other KYC images for multi-doc pipelines.
3. Tampered / forged samples showing overlays, copy-paste photos, font manipulations, print-scan artifacts.
4. Paired image + ground-truth text for OCR fine-tuning.
5. Metadata / logs for real-world KYC flows (e.g., device metadata, timestamp, geolocation as available and legally permitted).

4. Data Collection & Preprocessing (Module 1)

Collection plan

- *Download the Kaggle datasets mentioned above and inspect licensing and sample content. Prefer datasets that are labeled and have bounding boxes/annotations.*
- *Generate synthetic datasets to simulate edge cases (cropped images, low-light, rotation, scanned images, smartphone photos).*
- *If real PII-containing data is used, ensure it is stored in secure, access-controlled environments and masked/anonymized for model training.*

Preprocessing steps

- *Standardize image sizes and color spaces (convert to RGB, preserve aspect ratio, pad as needed).*
- *Image enhancement: denoising, contrast/brightness normalization, adaptive histogram equalization for low-light photos.*
- *Data augmentation: rotation, scale, perspective transforms, blurring, JPEG compression noise, copy-move tampering simulation for forgery class.*
- *Generate field-level crops using bounding boxes for OCR model fine-tuning.*
- *Create TFRecords / Parquet / Azure-friendly data blobs for efficient training in cloud.*

Labeling & annotation

- *Use labeling tools (LabelImg, VOTT, Roboflow) to create bounding boxes for fields and to annotate tampering regions.*
- *Define a consistent labeling schema (e.g., `name`, `aadhaar_number`, `dob`, `photo`, `address`, `tamper_mask`).*

5. AI-OCR & Document Verification (Module 2)

Architecture choices

- *Use Azure AI Document Intelligence (Form Recognizer) for initial field extraction — it provides pretrained models and custom model training on labeled fields.*
- *For local/custom OCR: Tesseract for baseline; fine-tune with synthetic fonts and domain-specific postprocessing.*
- *Combine visual layout models (LayoutLMv3-like architectures) for joint understanding of layout + text when available.*

Field extraction pipeline

1. *Use a document detector to locate the card within a photo (object detection model: Faster R-CNN / YOLOv8).*
2. *Correct perspective (homography) and crop to normalized card frame.*
3. *Run Azure Document Intelligence or a fine-tuned OCR engine on the cropped image to extract fields.*
4. *Apply rule-based validators: Aadhaar number checksum formats, DOB date parsing, name normalization, masked/unmasked formats.*
5. *Compute per-field confidence and an overall document confidence score.*

Fine-tuning with Azure OpenAI

- *Use Azure OpenAI to normalize extracted text (spell-correct names, standardize addresses, expand abbreviations) by prompting a lightweight text-cleaning model on extracted fields and their confidence scores.*

Cross-checks

- *Compare extracted photo with selfie (if provided) using face verification (face embeddings + threshold).*
- *Compare extracted Aadhaar number format using regex and masked vs. full presence rules.*

6. Fraud Detection using Deep Learning (Module 3)

Problem framing

- *Binary/multi-class classification: **genuine, tampered, low-quality, mismatch.***
- *Anomaly detection: unsupervised or semi-supervised models to detect out-of-distribution layouts or textures.*

Model ideas & architectures

- *Forgery Detection (image-level): CNN backbones (EfficientNet, ResNet) trained on tampered vs genuine images with focus on residual artifacts. Preprocessing can include high-pass or noise residual extraction to emphasize manipulation cues.*
- *Local tampering localization: U-Net / segmentation heads to highlight altered regions (copy-move detection).*
- *Siamese / metric-learning to compare document photo vs claimed selfie for person-match.*
- *Vision Transformer (ViT) or hybrid CNN+Transformer for global layout anomalies.*
- *Ensemble approach: Combine texture-based detectors (e.g., noiseprints) with deep models trained on simulated forgeries.*

Training strategy

- *Use balanced sampling across classes; oversample tampered cases via synthetic generation when real tampered examples are scarce.*
- *Loss functions: focal loss for class imbalance, dice loss for segmentation masks.*
- *Evaluation via cross-validation splits that keep source-device or source-batch separation to avoid overfitting to image acquisition artifacts.*

Real-time flagging

- *Models exported to ONNX and served with Azure Machine Learning Endpoints or Azure Container Instances for low-latency scoring.*
- *Set configurable thresholds to control precision/recall trade-offs depending on operational risk appetite.*

7. System Integration & Deployment (Module 4)

Integration points

- *REST API layer that accepts image + metadata; returns extracted fields, confidence scores, fraud-risk score, and evidence (region crops, annotated mask).*
- *Hook into existing KYC platforms via webhook or API.*

Azure deployment stack

- *Data storage: Azure Blob Storage for images and training artifacts.*
- *Model training: Azure ML or GPU-enabled compute instances.*
- *Inference: Azure Container Instances / Azure Kubernetes Service (AKS) for scale; Azure Cognitive Services for Document Intelligence calls.*
- *Logging & monitoring: Azure Application Insights and Azure Monitor for latency, error rates, and drift detection.*

User interface

- *Dashboard for operations teams to review flagged documents: show original image, cropped fields, OCR outputs, tamper mask overlay, and recommended action.*
- *Provide audit logs and evidence export for compliance.*

8. Evaluation, Metrics and Experiments

OCR evaluation

- *Field-level accuracy (exact match and normalized match), Character Error Rate (CER), Word Error Rate (WER).*
- *Per-field F1 score for detection/localization.*

Forgery detection evaluation

- *Precision, Recall, F1 for tampered vs genuine classification.*
- *AUROC for anomaly detectors.*
- *Intersection-over-Union (IoU) and pixel-wise F1 for segmentation/localization tasks.*

Operational metrics

- *End-to-end processing latency (ms).*
- *Human review rate (percentage of documents flagged for manual check).*
- *False positive rate and false negative risk analysis.*

Experiment plan (suggested)

1. *Baseline OCR using Tesseract on field crops — measure CER/WER.*
2. *Azure Document Intelligence custom model training with annotated samples — measure improvement over baseline.*
3. *Train forgery classifier on synthetic tamper dataset — evaluate F1/AUROC.*
4. *Test face verification module with public face datasets (non-sensitive) and paired selfie-document images to tune threshold.*
5. *End-to-end pipeline evaluation on a holdout set representing typical acquisition devices (mobile/scan).*

9. Privacy, Security, Legal & Ethical Considerations

- **Legal compliance:** *Aadhaar data is sensitive personal information and is protected under Aadhaar Act and data protection laws. Any handling of Aadhaar data must comply with UIDAI rules, relevant Indian laws, and any institutional review board (IRB) approvals. Avoid storing full Aadhaar numbers in plain text. Use masking and tokenization.*

- **Consent & minimal data:** Collect only what is strictly necessary and obtain explicit consent. Use masked Aadhaar or synthetic data wherever possible for model development.
 - **Data retention & access controls:** Encrypt data at rest and in transit. Implement role-based access and audit logs. Regularly purge data according to retention policy.
 - **Bias & fairness:** Evaluate model performance across demographic slices (age, gender, skin tones) to ensure fairness and avoid disproportionate false rejections.
 - **Explainability:** Provide explainable flags and visual evidence to support human review decisions.
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10. Implementation Roadmap & Resource Requirements

Revised Timeline — 2 Months (8 Weeks)

Week-by-Week Execution Plan

Week 1: Dataset Acquisition & Setup

- Download datasets (Kaggle Aadhaar dataset and others).
- Review licensing and ensure compliance.
- Create folder structure, storage buckets (Azure Blob), and data security setup.
- Begin initial annotation plan for fields (name, DOB, Aadhaar No., photo, address).

Week 2: Preprocessing & Synthetic Data Generation

- Apply image preprocessing steps (resize, denoise, normalize, deskew).
 - Create synthetic Aadhaar-like samples + tampered samples.
 - Create bounding box annotations using tools such as LabelImg/Roboflow.
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Week 3: Baseline OCR & Azure Document Intelligence Setup

- *Run baseline OCR experiments (Tesseract + OpenCV pipeline).*
- *Configure Azure Document Intelligence.*
- *Upload 50–100 samples to train a **Custom Document Model**.*
- *Evaluate field-level extraction accuracy.*

Week 4: OCR Fine-Tuning & Validation Rules Development

- *Fine-tune custom model with improved labeled samples.*
 - *Add rule-based checks: Aadhaar format validation, DOB parsing, address cleanup.*
 - *Integrate Azure OpenAI for text normalization.*
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Week 5: Fraud Detection Model (Deep Learning)

- *Build CNN/ViT-based classifier for tampered vs genuine.*
- *Train on genuine + synthetically tampered datasets.*
- *Begin development of tampering localization (U-Net segmentation).*

Week 6: Advanced Fraud Detection & Face Matching

- *Improve fraud classifier with noise residuals and texture-based features.*
 - *Add photo–selfie match module using Siamese network.*
 - *Test thresholds for low false positives.*
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Week 7: System Integration & API Development

- *Build REST API for document upload → OCR → fraud scoring.*
- *Integrate Azure services (OCR endpoint + ML endpoint).*
- *Build JSON response structure (fields, confidence, fraud score, evidence crops).*

Week 8: UI, Testing & Final Deployment

- *Build review dashboard for KYC team.*
 - *Test end-to-end pipeline with 100–200 documents.*
 - *Latency testing and optimization.*
 - *Prepare documentation, compliance checklist, final report.*
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11. Risks and Mitigations

. Risks and Mitigations

- **Legal risk:** *Do not use non-consented Aadhaar data. Mitigation: synthetic data and strict legal review.*
 - **Data scarcity for tampered examples:** *Use synthetic forgery generation and transfer learning from general forgery datasets.*
 - **Model overfitting to device artifacts:** *Use diverse acquisition devices and domain generalization techniques.*
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12. Conclusions and Future Work

This research document provides a structured plan to build an AI-driven Aadhaar identity verification and fraud prevention system with attention to data, modeling, deployment, and compliance. Future work includes on-device lightweight models for offline verification, continual learning for drift, and adversarial robustness testing.

Appendix A — Required Datasets & Acquisition Checklist

- [nagendra048/aadhar-dataset](#) (Kaggle) — field images for OCR and detection.
 - [nagendra048/pan-card-dataset](#) (Kaggle) — PAN card images for multi-doc support.
 - [varunkumargera/aadhar-images](#) (Kaggle) — alternative Aadhaar images.
 - [sparsh2002/govtiddataset](#) (Kaggle) — mixed govt ID images.
 - UIDAI offline eKYC sample data (UIDAI developer resources) for expected e-Aadhaar structure.
 - Synthetic tamper dataset: generate programmatically (copy-move, overlay, font changes, reprint-scan noise).
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Appendix B — Sample Data Schema (for training / inference blobs)

A typical data object conforms to the following JSON structure:

```
{
  "document_id": "uuid",
  "source": "kaggle|synthetic|user_upload",
  "image_uri": "https://...",
  "device_meta": {
    "device_type": "android",
    "app_version": "1.2.0"
  },
  "annotations": {
    "fields": [
      {
        "name": "name",
        "bbox": [x, y, w, h],
        "text": "..."
      }
    ],
    "tamper_mask_uri": "https://..."
  },
  "ocr_output": {
```

```
"name": {  
  "text": "...",  
  "confidence": 0.95  
},  
"fraud_score": 0.87,  
"created_at": " 2025-11-27 T00:00:00Z"  
}
```

Appendix C — Useful Tools & Libraries

- *Azure AI Document Intelligence (Form Recognizer)*
- *Azure OpenAI (text normalization and workflows)*
- *OpenCV, Pillow for image processing*
- *PyTorch / TensorFlow for model training*
- *Tesseract OCR for baseline*
- *Labellmg, VOTT, Roboflow for annotation*
- *ONNX for model export*

Document prepared as a research & implementation blueprint for prototyping an Aadhaar-based AI identity verification and fraud detection system.